

Smart Solar Charger



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Solar-powered lighting systems are now used in rural as well as urban areas. These systems include solar lanterns, solar home lighting systems, solar streetlights, solar garden lights, solar water heaters and solar power packs. All of them consist of four components: solar photovoltaic (PV) module, rechargeable battery, solar charge controller and load. The

solar charge controller plays an important role as the system's overall success depends mainly on it. It is considered as an indispensable link between the solar panel, battery and load.

Here we present the circuit of a highly efficient, automatic solar charger based on PIC16F877A microcontroller. It shows the system status on an LCD and can trickle charge.

Circuit and working

Fig. 1 shows the circuit of a microcontroller-based solar charger. In addition to microcontroller PIC16F877A (IC1), it uses regulator 7805 (IC2) and a few discrete components.

PIC16F877A is a powerful microcontroller that provides an ideal solution for hobby and industrial development. It controls battery charging through the solar panel. PIC microcontrollers use Harvard architecture.

PIC16F877A is an 8-bit, high-per-

formance RISC CPU with low power consumption. It has 8kB flash, 256 bytes of EEPROM, 368 bytes of RAM, 33 input/output (I/O) pins, 10-bit 8-channel analogue-to-digital converter (ADC), three timers, watchdog timer with its own on-chip R-C oscillator for reliable operation and synchronous FC interface. The microcontroller can recognise and execute only 35 simple instructions. All the instructions are single-cycle, except branches which are two-cycle instructions.

Port pins RB0 through RB7 of the microcontroller are connected to data pins D0 through D7 of the LCD module, respectively. Port pins RD1, RD2 and RD3 are connected to RS (register-select), R/W (read/write) and E (enable) of the LCD, respectively. Preset VR3 is used for contrast control. Switch S1 is used for manual reset. A 4MHz crystal along with two 33pF capacitors provides basic clock frequency to the

Test Points

Test point	Details
TP0	Ground (GND)
TP1	5V
TP2	12V (battery voltage)
TP3	3V (after calibration)
TP4	Solar panel voltage
TP5	3V (after calibration)
TP6	Input for current sense
TP7	Relay enable

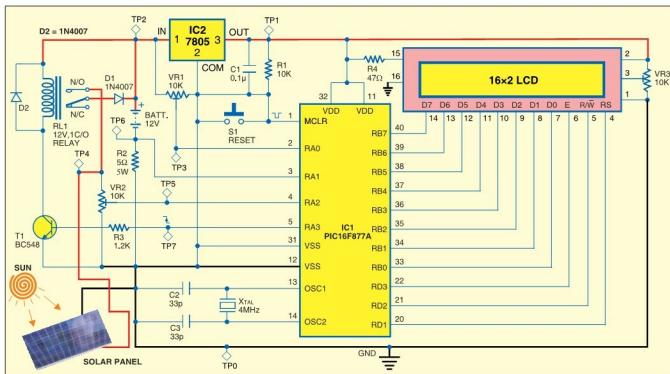


Fig. 1: Circuit of PIC16F877A microcontroller-based solar charger